

$$\int 2x \sqrt{x^2+1} dx = \int \sqrt{x^2+1} d\frac{x^2}{2} = \int \sqrt{x^2+1} d(x^2+1) = \int \frac{u - x^2}{\frac{d(x^2+1)}{dx} \cdot dx} =$$

$$= \int u^{\frac{1}{2}} du = \frac{u^{\frac{1}{2}+1}}{\frac{1}{2}+1} + C = \frac{2}{3} u^{\frac{3}{2}} + C = \frac{2}{3} (x^2+1)^{\frac{3}{2}} + C$$

✓ 1716

$$\int \frac{x^4 dx}{\sqrt{4+x^5}} = \int \frac{d\frac{x^5}{5}}{\sqrt{x^5+4}} = \frac{1}{5} \int \frac{d(x^5+4)}{\sqrt{x^5+4}} = \frac{1}{5} \int (x^5+4)^{-\frac{1}{2}} d(x^5+4) =$$

$$= \frac{1}{5} \cdot \frac{(x^5+4)^{-\frac{1}{2}+1}}{-\frac{1}{2}+1} + C = \frac{1}{5} \cdot \frac{2}{1} \sqrt{x^5+4} + C = \frac{2}{5} \sqrt{x^5+4} + C$$

✓ 1726

$$\int \frac{dx}{\cos^2 x \sqrt{1+\tan x}} = \int \frac{d \tan x}{\sqrt{1+\tan x}} = \int \frac{d(\tan x+1)}{\sqrt{1+\tan x}} = \frac{(1+\tan x)^{-\frac{1}{2}+1}}{-\frac{1}{2}+1} + C =$$

$$= 2 \sqrt{1+\tan x} + C$$

✓ 1734

$$\int e^x \sin(e^x) dx = \int \sin(e^x) de^x = -\cos e^x + C$$

✓ 1733

$$\int \left[\cos \left(2x - \frac{\pi}{4} \right) \right]^2 dx = \int \left[\cos \left(2x - \frac{\pi}{4} \right) \right]^2 d \left(2x - \frac{\pi}{4} \right) = \frac{1}{2} \int \left[\cos \left(2x - \frac{\pi}{4} \right) \right]^2 d \left(2x - \frac{\pi}{4} \right) =$$

$$= \frac{1}{2} \int \frac{d \left(2x - \frac{\pi}{4} \right)}{\cos^2 \left(2x - \frac{\pi}{4} \right)} = \frac{1}{2} \int d \tan \left(2x - \frac{\pi}{4} \right) = \frac{1}{2} \tan \left(2x - \frac{\pi}{4} \right) + C$$

✓ 1737

$$\int \frac{(2x-3)dx}{x^2-3x+8} = \int \frac{d \left(2 \frac{x^2}{2} - 3x \right)}{x^2-3x+8} = \int \frac{d(x^2-3x+8)}{x^2-3x+8} = \ln |x^2-3x+8| + C$$

✓ 1747

$$\int \cot(2x+1) dx = \frac{1}{2} \int \frac{\cos(2x+1)}{\sin(2x+1)} d(2x+1) = \frac{1}{2} \int \frac{d(\sin(2x+1))}{\sin(2x+1)} = \frac{1}{2} \ln |\sin(2x+1)| + C$$

✓ 1763

$$\frac{dx}{\sqrt{4-9x^2}} = \frac{1}{3} \int \frac{dx}{\sqrt{\frac{4}{9}-x^2}} = \frac{1}{3} \arcsin \frac{x \cdot 3}{2} + C = \frac{1}{3} \arcsin \frac{3x}{2} + C$$

$$\int \frac{dx}{2x^2+9} \stackrel{\sim 1762}{=} \frac{1}{2} \int \frac{dx}{x^2+\frac{9}{2}} = \frac{1}{2} \cdot \frac{1}{3/\sqrt{2}} \operatorname{arctg} \frac{x}{3/\sqrt{2}} + C = \frac{\sqrt{2}}{6} \operatorname{arctg} \frac{\sqrt{2}x}{3} + C$$

$$\int \frac{1+x}{11-x^2} dx \stackrel{\sim 1773}{=} \frac{1}{2} \int \frac{dx^2}{11-x^2} + \int \frac{dx}{11-x^2} = \frac{1}{2} \int \frac{d(1-x^2)}{11-x^2} + \operatorname{arcsinh} x =$$

$$\stackrel{\sim 1776}{=} -\frac{1}{2} \cdot \frac{(1-x^2)^{-\frac{1}{2}+1}}{-\frac{1}{2}+1} + \operatorname{arcsinh} x + C = -\sqrt{1-x^2} + \operatorname{arcsinh} x + C$$

$$\int \frac{x(1-x^2)}{1+x^4} dx \stackrel{\sim 1776}{=} \int \frac{x-x^3}{1+x^4} dx = \int \frac{x dx}{1+x^4} - \int \frac{x^3 dx}{1+x^4} = \frac{1}{2} \int \frac{dx^2}{1+x^4} - \frac{1}{4} \int \frac{d(x^4)}{1+x^4} =$$

$$\stackrel{\sim 1779}{=} \frac{1}{2} \operatorname{arctg} \frac{x^2}{1} - \frac{1}{4} \ln|x^4+1| + C = \frac{1}{2} \operatorname{arctg} x^2 - \frac{1}{4} \ln|x^4+1| + C$$

$$\int \frac{2x - \sqrt{\operatorname{arcsinh} x}}{\sqrt{1-x^2}} dx = \int \frac{2x dx}{\sqrt{1-x^2}} - \int \frac{\sqrt{\operatorname{arcsinh} x} dx}{\sqrt{1-x^2}} = \int \frac{d(1-x^2)}{\sqrt{1-x^2}} - \int \sqrt{\operatorname{arcsinh} x}.$$

$$\stackrel{\sim 1789}{=} d \operatorname{arcsinh} x = -\frac{(1-x^2)^{-\frac{1}{2}+1}}{-\frac{1}{2}+1} - \frac{(\operatorname{arcsinh} x)^{\frac{3}{2}+1}}{\frac{3}{2}+1} + C = -2\sqrt{1-x^2} - \frac{2}{3}(\operatorname{arcsinh} x)^{\frac{3}{2}} + C$$

$$\int \frac{x^4}{1-x} dx = - \int (x^3 + x^2 + x + \frac{1}{x-1} + 1) dx = - \int x^3 dx - \int x^2 dx - \int x dx -$$

$$- \int \frac{dx}{x-1} - \int dx = -\frac{x^4}{4} - \frac{x^3}{3} - \frac{x^2}{2} - x - \ln|x-1| + C$$

~ 1785

$$\int \frac{(2x-1)dx}{x-2} = \int \frac{(2x-2)+3}{x-2} dx = \int 2 dx + \int \frac{3}{x-2} dx = 2x + 3 \ln|x-2| + C$$